Solution Problem 7.1 (see also m-file)

CT SS

$$\frac{dx_1}{dt} = x_2 + w_1$$

$$\frac{dx_2}{dt} = \frac{1}{2} S_0 e^{-x_1/D} x_2^2 x_3 - g + w_2$$

$$\frac{dx_3}{dt} = 0 + w_3$$

$$z(t) = \sqrt{L^2 + (x_1(t) - a)^2} + v$$

DT SS

$$x_1(k) = x_1(k-1) + T x_2(k-1) + w_1(k-1)$$

$$x_2(k) = x_2(k-1) + \frac{T}{2} S_0 e^{-x_1(k-1)/D} x_2^2(k-1) x_3(k-1) - gT + w_2(k-1)$$

$$x_3(k) = x_3(k-1) + w_3(k-1)$$

$$z(k) = \sqrt{L^2 + (x_1(k) - a)^2} + v(k)$$

Jacobians for EKF

$$A_k = \left.\frac{\partial f_i}{\partial x_j}\right|_{\hat{x}_{k-1}} = \begin{bmatrix} 1 & -\frac{1}{D}\hat{x}_2^2(k-1)\hat{x}_3(k-1) \cdot \frac{T}{2} S_0 e^{-\hat{x}_1(k-1)/D} & 0 \\ T & 2\hat{x}_2(k-1)\hat{x}_3(k-1) \cdot \frac{T}{2} S_0 e^{-\hat{x}_1(k-1)/D} & 0 \\ 0 & \hat{x}_2^2(k-1) \cdot \frac{T}{2} S_0 e^{-\hat{x}_1(k-1)/D} & 1 \end{bmatrix}$$

(3×3)

$$W_k = \frac{\partial f_i}{\partial w_j} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

(3×3)

$$H_k = \left.\frac{\partial h_i}{\partial x_j}\right|_{\bar{x}_k} = \begin{bmatrix} \frac{\bar{x}_1(k) - a}{\sqrt{L^2 + (\bar{x}_1(k) - a)^2}} & 0 & 0 \end{bmatrix}$$

(1×3)

$$V_k = \frac{\partial h_i}{\partial w_j} = 1$$

(1×1)

Solution Problem 7.2 (see also m-file)

CT SS

$$\frac{dx}{dt} = -\frac{k_v}{A}\sqrt{\rho g\, x(t) - p_0} + \frac{1}{A}u(t) + w(t)$$

$$z(t) = x^2(t) + x^3(t) + v(t)$$

DT SS

$$x_k = x_{k-1} - \frac{k_v T_s}{A}\sqrt{\rho g\, x_{k-1} - p_0} + \frac{T_s}{A}u_{k-1} + w_{k-1}$$

$$z_k = x_k^2 + x_k^3 + v_k$$

Jacobians

$$A_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}} = 1 - \frac{k_v T_s \rho g}{2A}\frac{1}{\sqrt{\rho g\, \hat{x}_{k-1} - p_0}}$$

$$W_k = \frac{\partial f}{\partial w} = 1$$

$$H_k = \left.\frac{\partial z}{\partial x}\right|_{\hat{x}_k^-} = 2\hat{x}_k^- + 3(\hat{x}_k^-)^2$$

$$V_k = \frac{\partial z}{\partial v} = 1$$

EKF

$(\hat{x}_{k-1}, P_{k-1})$

Predict

$$\hat{x}_k^- = f(\hat{x}_{k-1}, u_{k-1}) + 0 = \hat{x}_{k-1} - \frac{k_v T_s}{A}\sqrt{\rho g\, \hat{x}_{k-1} - p_0} + \frac{T_s}{A}u_{k-1}$$

$$\hat{z}_k^- = (\hat{x}_k^-)^2 + (\hat{x}_k^-)^3$$

$$P_k^- = A_k^2 P_{k-1} + \sigma_w^2$$

$\hat{x}_k^-$, $\hat{z}_k^-$, P_k^-

$$K_k = \frac{P_k^- H_k}{H_k^2 P_k^- + \sigma_v^2}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - \hat{z}_k^-)$$

$$P_k = (1 - K_k H_k) P_k^-$$

$\hat{x}_k$, P_k

$k := k+1$

z_k

Update